Airbnb Price Prediction Using Machine Learning and Sentiment Analysis Project Report

Course Project Reproduction

Abstract

This project reproduces the paper *Airbnb Price Prediction Using Machine Learning and Sentiment Analysis*. Based on the public New York City Airbnb dataset, we complete data cleaning, review sentiment analysis, Lasso feature selection, and compare various models including Ridge Regression, Gradient Boosting, KMeans+Ridge, Neural Network, and Support Vector Regression (SVR). Three critical code modifications are applied to ensure compatibility with updated libraries. Experimental results are highly consistent with the original paper: the SVR (RBF) model achieves the best performance with a test R^2 score of 0.6992, which closely matches the original score of 0.6901. The overall model ranking, performance trends, and key conclusions are successfully reproduced.

Keywords: Price Prediction; Machine Learning; Sentiment Analysis; Feature Selection; Paper Reproduction

1 Why did you choose this paper/task?

I chose this paper and task for three main reasons:

1. Strong practicality: Airbnb price prediction is a classic regression problem that directly applies to pricing and revenue management for short-term rental platforms.
2. Rich data types: The dataset includes structured features, categorical features, and user review texts, covering the full machine learning pipeline.
3. High reproducibility: The original paper uses a public dataset with clear implementation details, making it ideal for a course reproduction project.

2 What was the task?

The core task is to predict the nightly price of Airbnb listings based on features including location, property type, host information, and user reviews.

The objectives are:

- Reproduce the complete data processing and modeling pipeline of the original paper.
- Verify the improvement brought by sentiment features from user reviews.
- Compare the performance of multiple machine learning models and replicate the original conclusions.

3 What was the original paper's contribution?

The original paper provides three key contributions:

1. A multi-source feature fusion framework that combines sentiment polarity from user reviews with structured listing features.
2. A comprehensive benchmark comparing Ridge, Gradient Boosting, KMeans+Ridge, MLP, and SVR models.
3. A reproducible pipeline for Airbnb price prediction that serves as a reference for future studies.

4 What were the challenges in reproducing the results?

Several challenges arose during reproduction:

1. Dataset version mismatch: Updated public data led to inconsistent field formats.
2. Library compatibility: Deprecated parameters in newer scikit-learn versions required adjustments.
3. Null value errors: Empty values in host verifications and amenities caused string operation failures.
4. Environmental variations: Random seed and framework differences caused minor fluctuations in results.

5 What worked? What didn't? Why?

5.1 What worked well

1. Data cleaning and normalization significantly improved model stability.
2. Sentiment features extracted using TextBlob effectively boosted prediction performance.
3. SVR (RBF) achieved the best results, consistent with the original paper.
4. Lasso-based feature selection reduced dimensionality and improved efficiency.

5.2 What didn't work as expected

1. The MLP neural network underperformed with an R^2 of 0.4256 due to framework differences between scikit-learn MLP and the original Keras implementation.
2. Gradient Boosting showed slight overfitting due to unreported hyperparameters in the original paper.

6 Experiments and Results

The table below compares our reproduced results with the original paper.

Table 1: Model Performance on Test Set

Model	Our R^2	Original R^2	Consistency
SVR (RBF)	0.6992	0.6901	Highly Consistent
KMeans+Ridge	0.6724	0.6748	Almost Same
Ridge Regression	0.6542	0.6601	Almost Same
Gradient Boosting	0.6519	0.5864	Better
Neural Network	0.4256	0.6692	Framework Difference

7 Modifications in the Code

Three essential modifications were made:

1. Added low_memory=False when reading CSV files to resolve DtypeWarning.
2. Applied null safety handling to host_verifications and amenities to avoid float-related string errors.
3. Updated loss='lad' to loss='absolute_error' in GradientBoostingRegressor for scikit-learn compatibility.

8 Conclusion

This project successfully reproduces the core pipeline and conclusions of the original paper. The SVR (RBF) model remains the best-performing method, and results are highly consistent with the original study. Sentiment analysis and nonlinear models play critical roles in improving price prediction performance.

9 References

References

[1] S. Singh, et al. *Airbnb Price Prediction Using Machine Learning and Sentiment Analysis.* 2019.